

Chương 4: Biến đổi Laplace và ứng dụng

Bài 4.1: Xét biến đổi Laplace, ROC và vị trí đ' không, vẽ miền của $X(s)$ của TH sau:

$$a) x(t) = e^{j\omega_0 t} u(t) = \begin{cases} e^{j\omega_0 t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$\Rightarrow \text{BD Laplace: } X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt$$

$$= \int_0^{+\infty} e^{j\omega_0 t} e^{-st} dt$$

$$= \int_0^{+\infty} e^{t(j\omega_0 - s)} dt$$

$$= \frac{1}{j\omega_0 - s} e^{t(j\omega_0 - s)} \Big|_{t=0}^{t=+\infty}$$

$$= \frac{1}{j\omega_0 - s} (e^{\infty(j\omega_0 - s)} - 1)$$

$$= -\frac{1}{j\omega_0 - s} + \frac{e^{\infty(j\omega_0 - s)}}{j\omega_0 - s}$$

$$\Rightarrow X(s) = \frac{1}{s - j\omega_0} \text{ khi } \operatorname{Re}(s) > 0.$$

hay $X(s) \rightarrow \infty$.

Điểm kỳ tại đ' hàm $\rightarrow \infty \Rightarrow X(s)$ có đ' của lư $s = j\omega_0$

$$(b) x(t) = \sin(3t) u(t) = \begin{cases} \sin 3t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$\Rightarrow$ BD Laplace:

$$X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt = \int_0^{+\infty} \sin 3t e^{-st} dt$$

$$= \int_0^{+\infty} \frac{1}{2j} (e^{j3t} - e^{-j3t}) e^{-st} dt$$

$$= \int_0^{+\infty} \left[\frac{1}{2j} e^{t(3j - s)} - \frac{1}{2j} e^{-t(3j + s)} \right] dt$$

$$= \frac{1}{2j(3j - s)} e^{t(3j - s)} \Big|_0^{+\infty} + \frac{1}{2j(3j + s)} e^{-t(3j + s)} \Big|_0^{+\infty}$$

$$= \frac{e^{\infty(3j-s)}}{2j(3j-s)} - \frac{1}{2j(3j-s)} + \frac{e^{-\infty(3j+s)}}{2j(3j+s)} - \frac{1}{2j(3j+s)}$$

$$= -\frac{1}{2j} \left(\frac{1}{3j-s} + \frac{1}{3j+s} \right) + \left(\frac{e^{\infty 3j - \infty s}}{2j(3j-s)} + \frac{e^{-\infty 3j - \infty s}}{2j(3j+s)} \right)$$

$$x(s) \text{ h\u00e1 tu } \Rightarrow \operatorname{Re}(s) > 0.$$

$$\Rightarrow x(s) = -\frac{1}{2j} \left(\frac{1}{3j-s} + \frac{1}{3j+s} \right) = \frac{3j+s+3j-s}{-2j(3j-s)(3j+s)} = \frac{6j}{-2j(9j^2-s^2)} = \frac{-3}{-9-s^2} = \frac{3}{s^2+9}$$

$$x(s) \rightarrow \infty \Leftrightarrow s = \pm j3.$$

$$\Rightarrow \text{V\u00e9l\u00e9 tai } s = 3j \text{ v\u00e0 } s = -3j.$$

$$\textcircled{c} \quad X(s) = \frac{2s+5}{s^2+5s+6} = \frac{2s+5}{(s+2)(s+3)} \quad \text{Sai}$$

$$\textcircled{c} \quad x(t) = e^{-2t} u(t) + e^{-3t} u(t).$$

$$e^{-2t} u(t) = \begin{cases} e^{-2t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$e^{-3t} u(t) = \begin{cases} e^{-3t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$\Rightarrow x(t) = \begin{cases} e^{-2t} + e^{-3t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$\Rightarrow$ BP Laplace.

$$X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt = \int_0^{+\infty} (e^{-2t} + e^{-3t}) e^{-st} dt.$$

$$= \int_0^{+\infty} (e^{-t(s+2)} + e^{-t(s+3)}) dt.$$

$$= \left. \frac{-1}{s+2} e^{-t(s+2)} \right|_0^{+\infty} + \left. \frac{-1}{s+3} e^{-t(s+3)} \right|_0^{+\infty}.$$

$$= \frac{-1}{s+2} (e^{-\infty(s+2)} - 1) + \frac{-1}{s+3} (e^{-\infty(s+3)} - 1).$$

$$= \frac{1}{s+2} + \frac{1}{s+3} - \left(\frac{e^{-\infty(s+2)}}{s+2} + \frac{e^{-\infty(s+3)}}{s+3} \right).$$

$$= \frac{2s+5}{s^2+5s+6} - \left(\frac{e^{-2(s+2)}}{s+2} + \frac{e^{-(s+3)}}{s+3} \right)$$

$$\Rightarrow X(s) = \frac{2s+5}{s^2+5s+6}$$

$$ROC: \begin{cases} \operatorname{Re}(s) + 2 > 0 \\ \operatorname{Re}(s) + 3 > 0 \end{cases}$$

$$\Rightarrow \begin{cases} \operatorname{Re}(s) > -2 \\ \operatorname{Re}(s) > -3 \end{cases}$$

$$\Rightarrow \operatorname{Re}(s) > -2$$

$$X(s) = 0 \Rightarrow s = -\frac{5}{2}$$

$$X(s) \rightarrow \infty \Rightarrow s = -2, s = -3$$

Vậy đ không tại $s = -\frac{5}{2}$

đ của tại $s = -2; s = -3$

Bài 4.2: Tìm biến đổi Laplace 1 phía của TH sau:

(a) $x(t) = e^{-t}(t-2)u(t-2)$

$$= e^{-t}(t-2) \quad t \geq 2$$

$$0 \quad t < 2$$

$\Rightarrow$ BT Laplace 1 phía

$$X(s) = \mathcal{L}\{x(t)\} = \int_0^{\infty} x(t)e^{-st} dt$$

$$= \int_0^{\infty} e^{-t}(t-2)u(t-2)e^{-st} dt$$

$$= \int_2^{\infty} e^{-t}(t-2)e^{-st} dt$$

$$= \int_2^{\infty} e^{-t(st+1)}(t-2) dt$$

$$\begin{cases} u = t-2 \\ du = 1 dt \end{cases} \Rightarrow \begin{cases} du = 1 dt \\ u = -\frac{1}{s+1} e^{-(s+1)t} \end{cases}$$

$$\begin{aligned} \Rightarrow X(s) &= -\frac{(t-2)}{s+1} e^{-(s+1)t} \Big|_2^{\infty} + \int_2^{\infty} \frac{e^{-(s+1)t}}{s+1} dt \\ &= -\frac{(t-2)e^{-(s+1)t}}{s+1} \Big|_2^{\infty} - \frac{e^{-(s+1)t}}{(s+1)^2} \Big|_2^{\infty} \end{aligned}$$

$$= \frac{e^{-\infty(s+1)}}{(s+1)^2} + \frac{e^{-2(s+1)}}{(s+1)^2} - \frac{\infty \cdot e^{-\infty(s+1)}}{s+1} + 0.$$

$$\Rightarrow X(s) = \frac{e^{-2(s+1)}}{(s+1)^2}.$$

* (b). $x(t) = t^2 e^{-2t} u(t) = \begin{cases} t^2 e^{-2t} & t \geq 0 \\ 0 & t < 0. \end{cases}$

$\Rightarrow$ BD Laplace 1 phia:

$$X^1(s) = \int_0^{\infty} t^2 e^{-2t} dt = \int_0^{\infty} t^2 e^{-t(s+2)} dt.$$

$$\begin{cases} u = t^2 \\ du = 2t dt \\ \int du = e^{-t(s+2)} \end{cases} \Rightarrow v = -\frac{1}{2} e^{-t(s+2)}.$$

$$\Rightarrow X^1(s) = \frac{-t^2 e^{-t(s+2)}}{2} \Big|_0^{\infty} + \int_0^{\infty} \frac{2t e^{-t(s+2)}}{2} dt.$$

$$X^1(s) = \frac{-t^2 e^{-t(s+2)}}{2} \Big|_0^{\infty} + \int_0^{\infty} \frac{2t e^{-t(s+2)}}{2} dt.$$

$$= \frac{-t^2 e^{-t(s+2)}}{2} \Big|_0^{\infty} + \frac{1}{2}.$$

$$I_2 = \int_0^{\infty} t e^{-t(s+2)} dt.$$

$$\begin{cases} u = t \\ du = dt \\ dv = e^{-t(s+2)} \end{cases} \Rightarrow v = -\frac{1}{s+2} e^{-t(s+2)}$$

$$\Rightarrow I_2 = \frac{-t e^{-t(s+2)}}{s+2} \Big|_0^{\infty} + \int_0^{\infty} \frac{1}{s+2} e^{-t(s+2)} dt$$

$$= \frac{-t e^{-t(s+2)}}{s+2} \Big|_0^{\infty} - \frac{1}{(s+2)^2} e^{-t(s+2)} \Big|_0^{\infty}$$

$$\Rightarrow X^1(s) = \frac{-\infty e^{-\infty(s+2)}}{2} + \frac{-\infty e^{-\infty(s+2)}}{s+2} - \frac{e^{-\infty(s+2)}}{(s+2)^2} + \frac{1}{(s+2)^2}.$$

$$c) \cdot x(t) = t u(t) - (t-1) u(t-1) - (t-2) u(t-2) + (t-3) u(t-3).$$

$$t u(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$(t-1) u(t-1) =$$

$$X^1(s) = \int_{-\infty}^{\infty} t e^{-st} dt - \int_{-\infty}^{\infty} (t-1) e^{-st} dt - \int_{-\infty}^{\infty} (t-2) e^{-st} dt + \int_{-\infty}^{\infty} (t-3) e^{-st} dt.$$

$$\int_{-\infty}^{\infty} t e^{-st} dt = -\frac{t e^{-st}}{s} \Big|_0^{\infty} + \int_0^{\infty} \frac{1}{s} e^{-st} dt = \int_0^{\infty} \sin t dt$$

$$I = \int_a^{\infty} (t-a) e^{-st} dt.$$

$$\begin{cases} u = t-a \\ dv = e^{-st} dt \end{cases} \Rightarrow \begin{cases} du = dt \\ v = -\frac{1}{s} e^{-st} \end{cases}$$

$$\Rightarrow I = -\frac{(t-a) e^{-st}}{s} \Big|_a^{\infty} + \int_a^{\infty} \frac{e^{-st}}{s} dt.$$

$$= -\frac{(t-a) e^{-st}}{s} \Big|_a^{\infty} - \frac{1}{s^2} e^{-st} \Big|_a^{\infty}$$

$$= \frac{e^{-\infty}}{s} - \frac{1}{s^2} e^{-\infty} + \frac{1}{s^2} e^{-sa}.$$

$$\Rightarrow X^1(s) = \frac{1}{s^2} (e^{-0s} - e^{-s} - e^{-2s} - e^{-3s}) = \frac{1}{s^2} (1 - e^{-s} - e^{-2s} - e^{-3s})$$

$$d) \cdot x(t) = e^{-t} u(t) + \cos(t-2) u(t-2).$$

$$X^1(s) = X_1^1(s) X_2^2(s).$$

$$\cdot x_1(t) = e^{-t} u(t) \Rightarrow X_1^1(s) = \int_0^{\infty} e^{-t} e^{-st} dt = \int_0^{\infty} e^{-(s+1)t} dt.$$

$$= -\frac{1}{s+1} e^{-(s+1)t} \Big|_0^{\infty} = -\frac{e^{-\infty(s+1)}}{s+1} + \frac{1}{s+1}.$$

$$\Rightarrow X_1^1(s) = \frac{1}{s+1}.$$

$$x_2(t) = \cos(t-2)u(t-2) = \frac{1}{2}(e^{j(t-2)} + e^{-j(t-2)}) \quad t \geq 2.$$

$$\Rightarrow X_2^1(s) = \frac{1}{2} \left[\int_2^\infty (e^{j(t-2)} e^{-st} + e^{-j(t-2)} e^{-st}) dt \right]$$

$$= \frac{1}{2} \left[\int_2^\infty (e^{(j-s)t-2j} + e^{-(j+s)t+2j}) dt \right]$$

$$= \frac{1}{2} \left[\int_2^\infty e^{-(s-j)t} dt + \int_2^\infty e^{-(s+j)t} dt \right] e^{2j}$$

$$= \frac{e^{2j}}{2} \left[\frac{1}{s-j} e^{-(s-j)t} \Big|_2^\infty - \frac{1}{s+j} e^{-(s+j)t} \Big|_2^\infty \right]$$

$$= \frac{e^{2j}}{2} \cdot \frac{e^{-2(s-j)}}{s-j} + \frac{e^{2j}}{2} \cdot \frac{e^{-2(s+j)}}{s+j} - \frac{e^{2j} e^{-2(s-j)}}{2(s-j)} - \frac{e^{2j} e^{-2(s+j)}}{2(s+j)}$$

$$X_2^1(s) = \frac{e^{-2s}}{2(s-j)} + \frac{e^{-2s}}{2(s+j)}$$

$$= \frac{e^{-2s} (s+j+s-j)}{2(s-j)(s+j)} = \frac{e^{-2s} \cdot 2s}{2(s^2-j^2)}$$

$$= \frac{e^{-2s} \cdot s}{s^2+1}$$

$$\Rightarrow X^1(s) = \frac{e^{-2s} \cdot s}{(s+1)(s^2+1)}$$

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Bài 4.3, Tìm giá trị đầu và giá trị cuối của $x(t)$ ứng với:

$$(a) X(s) = e^{-5s} \frac{-2}{s(s+2)}$$

$$\text{Giá trị đầu } x(0) = \lim_{s \rightarrow \infty} s e^{-5s} \frac{-2}{s(s+2)} = \lim_{s \rightarrow \infty} \frac{-2 \cdot e^{-5s}}{s+2} = 0.$$

$$\text{Giá trị cuối } x(\infty) = \lim_{s \rightarrow 0} s e^{-5s} \frac{-2}{s(s+2)}$$

$$= \lim_{s \rightarrow 0} \frac{-2e^{-5s}}{s+2} = -1.$$

$$b) X(s) = \frac{2s+3}{s^2+5s+6}.$$

$$\text{Giá trị đầu } x(0) = \lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} \frac{2s^2+3s}{s^2+5s+6} = 2.$$

$$\text{Giá trị cuối } x(\infty) = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} \frac{2s^2+3s}{s^2+5s+6} = \frac{0}{6} = 0.$$

Bài 4.4 Tìm Laplace ngược;

$$a) X(s) = \frac{-5s-7}{(s+1)(s-1)(s+2)}.$$

$$= \frac{A}{s+1} + \frac{B}{s-1} + \frac{C}{s+2}.$$

$$\Rightarrow A(s-1)(s+2) + B(s+1)(s+2) + C(s+1)(s-1) = -5s-7.$$

$$\Rightarrow AS^2 + 2AS - 2A + BS^2 + 3BS + 2B + CS^2 - C = -5s-7.$$

$$\Rightarrow \begin{cases} A+B+C=0 \\ 2A+3B-C=-5 \\ -2A+2B-C=-7 \end{cases}$$

$$\Rightarrow \begin{cases} A=1 \\ B=-2 \\ C=1 \end{cases}$$

$$\Rightarrow X(s) = \frac{1}{s+1} - \frac{2}{s-1} + \frac{1}{s+2}.$$

$$L^{-1} \left[\frac{1}{s+1} \right] = \begin{cases} e^{-t} u(t) & \text{Re} > -1 \\ -e^{-t} u(-t) & \text{Re} < -1 \end{cases}$$

$$L^{-1} \left[\frac{-2}{s-1} \right] = \begin{cases} -2e^t u(t) & \text{Re} > 1 \\ 2e^t u(-t) & \text{Re} < 1 \end{cases}$$

$$L^{-1} \left[\frac{1}{s+2} \right] = \begin{cases} e^{-2t} u(t) & \text{Re} > -2 \\ -e^{-2t} u(-t) & \text{Re} < -2 \end{cases}$$

$$\Rightarrow L^{-1}[X(s)] = \begin{cases} e^{-t} - 2e^{-t}u(t) + e^{-2t}u(t) & \operatorname{Re}(s) > 1 \\ -e^{-t}u(-t) + 2e^{-t}u(-t) - e^{-2t}u(-t) & \operatorname{Re}(s) < -2 \\ e^{-t}u(t) + 2e^{-t}u(-t) + e^{-2t}u(t) & -1 < \operatorname{Re}(s) < 1 \end{cases}$$

$$\Rightarrow x(t) = (e^{-t} - 2e^{-t} + e^{-2t})u(t).$$

$$(b) X(s) = \frac{s}{s^2 + 5s + 6} = \frac{A}{s+2} + \frac{B}{s+3}.$$

$$\Rightarrow 2B + 3A + AS = S.$$

$$\Rightarrow \begin{cases} B + A = 1 \\ 2B + 3A = 0 \end{cases} \Rightarrow \begin{cases} A = -2 \\ B = 3 \end{cases}$$

$$\Rightarrow X(s) = \frac{-2}{s+2} + \frac{3}{s+3}.$$

$$L^{-1}\left[\frac{-2}{s+2}\right] = \begin{cases} -2e^{-2t}u(t) & \operatorname{Re}(s) > -2 \\ +2e^{-2t}u(-t) & \operatorname{Re}(s) < -2 \end{cases}$$

$$L^{-1}\left[\frac{3}{s+3}\right] = \begin{cases} 3e^{-3t}u(t) & \operatorname{Re}(s) > -3 \\ -3e^{-3t}u(-t) & \operatorname{Re}(s) < -3 \end{cases}$$

$$\Rightarrow L^{-1}[X(s)] = \begin{cases} -2e^{-2t}u(t) + 3e^{-3t}u(t) & \operatorname{Re}(s) > -3 \\ 2e^{-2t}u(-t) - 3e^{-3t}u(-t) & \operatorname{Re}(s) < -2 \\ -2e^{-2t}u(t) - 3e^{-3t}u(-t) & -2 < \operatorname{Re}(s) < -3 \end{cases}$$

$$\Rightarrow x(t) = (-2e^{-2t}u(t) + 3e^{-3t})u(t).$$

$$(c) X(s) = \frac{s^2 + s - 3}{s^2 + 3s + 2} = \left(\frac{A}{s+1} + \frac{AB}{s+2} + 4\right) \operatorname{sai}$$

$$= 1 + \frac{-2s-5}{(s+1)(s+2)} = 1 + \frac{A}{s+1} + \frac{B}{s+2}.$$

$$\Rightarrow AS + 2A + BS + B = -2s - 5$$

$$\Rightarrow \begin{cases} A + B = -2 \\ 2A + B = -5 \end{cases} \Rightarrow \begin{cases} A = -3 \\ B = 1 \end{cases}$$

$$\Rightarrow X(s) = 1 - \frac{3}{s+1} + \frac{1}{s+2}.$$

$$\mathcal{L}^{-1}[1] = \delta(t)$$

$$\mathcal{L}^{-1}\left[\frac{-3}{s+1}\right] = \begin{cases} -3e^{-t}u(t) & \text{Re}(s) > -1 \\ +3e^{-t}u(-t) & \text{Re}(s) < -1 \end{cases}$$

$$\mathcal{L}^{-1}\left[\frac{+1}{s+2}\right] = \begin{cases} +1e^{-2t}u(t) & \text{Re}(s) > -2 \\ -e^{-2t}u(-t) & \text{Re}(s) < -2 \end{cases}$$

$$\Rightarrow \mathcal{L}^{-1}[X(s)] = \begin{cases} \delta(t) - 3e^{-t}u(t) + e^{-2t}u(t) & \text{Re}(s) > -1 \\ \delta(t) - 3e^{-t}u(t) + e^{-2t}u(t) & -2 < \text{Re}(s) < -1 \\ \delta(t) + 3e^{-t}u(-t) - e^{-2t}u(-t) & \text{Re}(s) < -2 \end{cases}$$

$$\Rightarrow x(t) = \delta(t) + (-3e^{-t}u(t) + e^{-2t}u(t))$$

① $X(s) = \frac{4s^2+6}{s^2+s^3-2}$ ROC: $-1 < \text{Re}(s) < 1$

$$= \frac{A}{s-1} + \frac{Bs+C}{(s+1)^2+1}$$

$$\Rightarrow A(s^2+2s+2) + (Bs+C)(s-1) = 4s^2+6$$

$$\Rightarrow As^2 + 2AS + 2A + Bs^2 + sC - BC - C = 4s^2 + 6$$

$$\Rightarrow \begin{cases} A+B=4 \\ 2A-B+C=0 \\ 2A-C=6 \end{cases} \Rightarrow \begin{cases} A=2 \\ B=2 \\ C=-2 \end{cases}$$

$$\Rightarrow X(s) = \frac{2}{s-1} + \frac{2s-2}{(s+1)^2+1} = \frac{2}{s-1} + \frac{2s+2-4}{(s+1)^2+1}$$

$$= \frac{2}{s-1} + 2 \cdot \frac{s+1}{(s+1)^2+1} - 4 \cdot \frac{1}{(s+1)^2+1}$$

$$\bullet \mathcal{L}^{-1}\left[\frac{2}{s-1}\right] = \begin{cases} 2e^t u(t) & \text{Re}(s) > 1 \\ -2e^t u(-t) & \text{Re}(s) < 1 \end{cases} \Rightarrow x(t) = -2e^t u(-t)$$

$$\bullet \mathcal{L}^{-1}\left[\frac{B_1(s-a)}{(s-a)^2+\omega_0^2}\right] = B_1 e^{at} \cos(\omega_0 t) u(t)$$

$$\Rightarrow \mathcal{L}^{-1}\left[2 \cdot \frac{s+1}{(s+1)^2+1}\right] = 2e^{-t} \cos(t) u(t)$$

$$\bullet \mathcal{L}^{-1}\left[\frac{B_2 \omega_0}{(s-a)^2+\omega_0^2}\right] = B_2 e^{at} \sin(\omega_0 t) u(t) \Rightarrow \mathcal{L}^{-1}\left[\frac{-4 \cdot 1}{(s+1)^2+1}\right] = -4e^{-t} \sin(t) u(t)$$

PT: $(s-a)^2 + \omega_0^2 \cos \theta_0$

$s = a + j\omega_0$

~~$2sa + \omega_0^2 + a^2$~~

~~a^2~~

Date:

$$\text{Clm}_1: L^{-1} \left[\frac{s-a}{(s-a)^2 + (\omega_0)^2} \right] = e^{at} \cos(\omega_0 t) u(t).$$

$$= L^{-1} \cdot \frac{1}{2} \left[\frac{s - (a - j\omega_0) + s - (a + j\omega_0)}{[s - (a - j\omega_0)][s - (a + j\omega_0)]} \right]$$

$$= L^{-1} \cdot \frac{1}{2} \left[\frac{1}{s - (a + j\omega_0)} + \frac{1}{s - (a - j\omega_0)} \right].$$

$$= \frac{1}{2} (e^{(a+j\omega_0)t} + e^{(a-j\omega_0)t}) u(t).$$

$$= \frac{1}{2} e^{at} (e^{j\omega_0 t} + e^{-j\omega_0 t}) u(t).$$

$$= e^{at} \cos(\omega_0 t) u(t).$$

$$\text{Clm}_2: L^{-1} \left[\frac{\omega_0}{(s-a)^2 + (\omega_0)^2} \right] = e^{at} \sin(\omega_0 t) u(t).$$

$$= L^{-1} \left[\frac{1}{2j} \frac{2j\omega_0}{(s-a)^2 + \omega_0^2} \right] u(t).$$

$$= \frac{1}{2j} L^{-1} \left[\frac{s - (a - j\omega_0) - (s - (a + j\omega_0))}{(s - (a + j\omega_0))(s - (a - j\omega_0))} \right] u(t)$$

$$= \frac{1}{2j} L^{-1} \left[\frac{1}{s - (a + j\omega_0)} - \frac{1}{s - (a - j\omega_0)} \right] u(t)$$

$$= \frac{1}{2j} [e^{(a+j\omega_0)t} - e^{(a-j\omega_0)t}] u(t)$$

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$$= e^{at} \frac{1}{2} (e^{j\omega t} - e^{-j\omega t}) u(t)$$

$$= e^{at} \sin(\omega t) u(t)$$

Đai 4.5

$$(a) \frac{dy(t)}{dt} + 3y(t) = 4x(t)$$

$$x(t) = \cos(2t) u(t) = \frac{1}{2} e^{j2t} + \frac{1}{2} e^{-j2t}$$

$$y(0) = -2 \Rightarrow \frac{1}{2} \cdot \frac{1}{s-j2} + \frac{1}{2} \cdot \frac{1}{s+j2} = X(s)$$

$$\text{Laplace 1 phía} \Rightarrow sY(s) - y(0) + 3Y(s) = 4X(s)$$

$$\Rightarrow Y(s)(s+3) = 4X(s) + y(0)$$

$$\Rightarrow Y(s) = \frac{y(0)}{s+3} + \frac{4X(s)}{s+3} = \frac{-2}{s+3} + 4 \frac{X(s)}{s+3}$$

$$= Y_0(s) + Y_1(s)$$

$$Y_0(s) = \frac{-2}{s+3} \Rightarrow y_0(t) = \mathcal{L}^{-1}[Y_0(s)] = -2e^{-3t} u(t)$$

$$Y_1(s) = \frac{4X(s)}{s+3}$$

$$x(t) = \cos(2t) u(t) = \frac{1}{2} (e^{j2t} + e^{-j2t}) \quad t \geq 0$$

$$X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt \quad t < 0$$

$$= \int_0^{+\infty} \frac{1}{2} (e^{j2t} e^{-st} + e^{-j2t} e^{-st}) dt$$

$$= \frac{1}{2} \int_0^{+\infty} (e^{-(s-j2)t} + e^{-(s+j2)t}) dt$$

$$= \frac{1}{2} \left[\frac{1}{s-j2} e^{-(s-j2)t} + \frac{1}{s+j2} e^{-(s+j2)t} \right] \Big|_0^{+\infty}$$

$$= \frac{-1}{2(s-j2)} + \frac{1}{2(s+j2)} + \frac{e^{\infty(s-j2)}}{2(s-j2)} - \frac{e^{-\infty(s+j2)}}{2(s+j2)}$$

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$$= \frac{-1}{2} \left(\frac{1}{s-3} - \frac{1}{s+3} \right) = \frac{-1}{2} \left(\frac{2s+3+2s+3}{s^2-9} \right)$$

$$= \frac{+4s}{s^2+4}$$

$$\Rightarrow Y(s) = \frac{+4s}{(s+3)(s^2+4)}$$

$$= \frac{A}{s+3} + \frac{Bs+C}{s^2+4}$$

$$\Rightarrow As^2+4A+(Bs+C)(s+3) = +4s$$

$$\Rightarrow As^2+4A+Bs^2+Cs+3Bs+3C = -4s$$

$$\Rightarrow \begin{cases} A+B=0 \\ C+3B=-4 \\ 4A+3C=0 \end{cases} \Rightarrow \begin{cases} A = \frac{12}{13} \\ B = +\frac{12}{13} \\ C = \frac{16}{13} \end{cases}$$

$$\Rightarrow Y(s) = \frac{12}{13} \cdot \frac{1}{s+3} + \frac{\frac{12}{13}s + \frac{16}{13}}{s^2+4} = \frac{12s}{13(s^2+4)} + \frac{8}{13} \cdot \frac{2}{s^2+4}$$

$$\mathcal{L}^{-1} \left[\frac{12}{13} \frac{1}{s+3} \right] = \frac{12}{13} e^{-3t} u(t)$$

$$\mathcal{L}^{-1} \left[\frac{12}{13} \frac{s}{s^2+4} \right] = \frac{12}{13} e^{0t} \cos 2t = \frac{12}{13} \cos 2t$$

$$\mathcal{L}^{-1} \left[\frac{8}{13} \frac{2}{s^2+4} \right] = \frac{8}{13} e^{0t} \sin 2t = \frac{8}{13} \sin 2t$$

$$\Rightarrow Y(s(t)) = \frac{12}{13} e^{-3t} u(t) + \frac{12}{13} \cos 2t + \frac{8}{13} \sin 2t \quad \text{PHÚC THỊNH}$$

$$(b). \frac{d^2 y(t)}{dt^2} + 4y(t) = 8x(t).$$

$$\begin{cases} x(t) = u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \Rightarrow X(s) = \frac{1}{s} \\ y(0) = 1 \\ \left. \frac{d}{dt} y(t) \right|_{t=0} = 2 \end{cases}$$

Laplace 1 phía 2 vé

$$\Rightarrow (s^2 y(s) - s y(0) - \left. \frac{dy(t)}{dt} \right|_{t=0}) + 4y(s) = 8X(s).$$

$$\Rightarrow s^2 y(s) - s - 2 + 4y(s) = 8X(s).$$

$$\Rightarrow y(s) (s^2 + 4) - (s + 2) = 8X(s)$$

$$\Rightarrow y(s) = \frac{8X(s)}{s^2 + 4} + \frac{s + 2}{s^2 + 4} = \frac{8X(s) + s}{s^2 + 4} + \frac{2}{s^2 + 4}$$

$$= Y_s(s) + Y_0(s)$$

$$Y_0(s) = \frac{s + 2}{s^2 + 4} \Rightarrow y_0(t) = L^{-1}[Y_0(s)] = L^{-1}\left[\frac{s + 2}{s^2 + 4}\right] = e^0 \cos 2t + e^0 \sin 2t$$

$$\Rightarrow y_0(t) = \cos(2t)u(t) + \sin(2t)u(t).$$

$$Y_s(s) = \frac{8X(s)}{s^2 + 4} = \frac{8}{s(s^2 + 4)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 4}$$

$$\Rightarrow As^2 + 4A + Bs^2 + Cs = 8.$$

$$\Rightarrow A + B = 0.$$

$$4A = 8.$$

$$C = 0.$$

$$\Rightarrow \begin{cases} A = 2 \\ B = -2 \\ C = 0 \end{cases}$$

$$B = -2.$$

$$C = 0.$$

$$\Rightarrow Y_s(s) = \frac{2}{s} - \frac{2s}{s^2 + 4}.$$

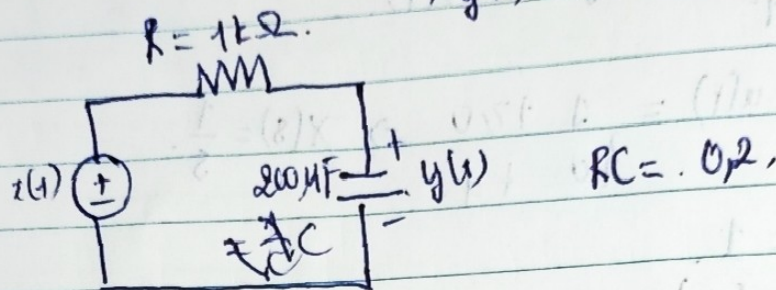
$$L^{-1}\left[\frac{2}{s}\right] = 2e^0 u(t) = 2u(t).$$

$$L^{-1}\left[\frac{-2s}{s^2 + 4}\right] = -2\cos(2t)u(t).$$

$$\Rightarrow y_s(t) = L^{-1}[Y_s(s)] = 2u(t) - 2\cos(2t)u(t).$$

Genios

Giải thích 4.6. (1) $\Rightarrow y(t) + \frac{1}{RC} \int_{-\infty}^t y(t) dt = \frac{1}{R} x(t)$
 $\Rightarrow \frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{R} \frac{dx(t)}{dt} = \frac{1}{RC} x(t)$
 Bài 4.6. Cho mạch sau và $x(t) = \left(\frac{3}{5}\right) e^{-2t} u(t)$
 $y(0) = -2$



$R: v(t) = Ri(t)$

$C: v(t) = \frac{1}{C} \int_{-\infty}^t i(t) dt$ và $y_s(t)$

Giải

Tác động: $v(t)_R + v(t)_C = v(t) \Rightarrow Ri(t) + \frac{1}{C} \int_{-\infty}^t i(t) dt = v(t)$

$\Rightarrow Ry(t) + \frac{1}{C} \int_{-\infty}^t y(t) dt = x(t)$ (1)

$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$

Laplace 1 phía 2 chiều

$\Rightarrow Sy'(s) + y(0) + \frac{1}{RC} y'(s) = \frac{1}{RC} X'(s)$

$\Rightarrow Sy'(s) + 2 + \frac{1}{0.2} y'(s) = \frac{1}{0.2} X_1(s)$

$\Rightarrow sy'(s) + 2 + 5y'(s) = 5X_1(s)$

$\Rightarrow y'(s) = \frac{1}{s+5} [5X(s) - 2] = \frac{-2}{s+5} + \frac{5X(s)}{s+5} = Y_0'(s) + Y_s(s)$

$x(t) = \frac{3}{5} e^{-2t} u(t) \Rightarrow X'(s) = \frac{3}{5} \int_0^{\infty} e^{-2t} e^{-st} dt$

$= \frac{3}{5} \int_0^{\infty} \frac{1}{2+s} e^{-t(2+s)} dt$

$= \frac{3}{5} \cdot \frac{1}{2+s} = \frac{3}{5(2+s)}$

$Y_0'(s) = \frac{-2}{s+5} \Rightarrow L^{-1}[Y_0'(s)] = y_0(t)$

$= -2 \cdot e^{-5t} u(t)$

$Y_s'(s) = \frac{3}{(s+2)(s+5)} = \frac{A}{s+2} + \frac{B}{s+5}$

$\Rightarrow \begin{cases} A+B=0 \\ 5A+2B=3 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \end{cases}$

$\Rightarrow Y_s'(s) = \frac{1}{s+2} - \frac{1}{s+5} \Rightarrow y_s(t) = L^{-1}[Y_s'(s)] = (e^{-2t} - e^{-5t}) u(t)$

Bài 4.7. Xét hàm truyền sau:

$$a) H(s) = \frac{s^2 - 2s - 3}{(s+2)(s^2 + 4s + 5)} = \frac{s^2 - 2s - 3}{(s+2)(s^2 + 4s + 5)} = \frac{(s^2 - 2s - 3)}{(s+2)(s+2)^2 + 1}$$

Đặc của hệ thống: $s = -2$; $s = -2 - i$; $s = -2 + i$ $\left| \begin{array}{l} \text{Re}(s) < 0 \end{array} \right. \Rightarrow$ Hệ thống ổn định

Xét tính nhân quả

$$As^2 + 4As + 5A + Bs^2 + 2Bs + C + 2C = s^2 - 2s - 3$$

$$\Rightarrow \begin{cases} A+B=1 \\ 4A+2B+C=-2 \\ 5A+2C=-3 \end{cases} \Rightarrow \begin{cases} A=5 \\ B=-4 \\ C=-14 \end{cases}$$

$$\Rightarrow H(s) = \frac{5}{s+2} - \frac{4s+4}{(s+2)^2+1} = \frac{5}{s+2} - \frac{4(s+2) + 6}{(s+2)^2+1}$$

$$\Rightarrow h(t) = 5e^{-2t} u(t) - (4e^{-2t} \cos t + 6e^{-2t} \sin t) u(t)$$

$\Rightarrow$ Hệ nhân quả:

$$b) H(s) = \frac{s^2 + 2s + 1}{(s^2 + 3s + 2)(s^2 + s - 2)}$$

$$= \frac{(s+1)^2}{(s+1)(s+2)(s-1)(s+2)}$$

$$= \frac{s+1}{(s+2)(s-1)(s+2)} = \frac{s+1}{(s-1)(s+2)^2}$$

$$= \frac{A}{s+2} + \frac{B}{s-1} + \frac{C}{(s+2)^2}$$

$$\Rightarrow A(s-1)(s+2) + B(s+2)^2 + C(s-1) = s+1$$

$$\Rightarrow (s-1)(As+2A) + Bs^2 + 4Bs + 4B + Cs - C = s+1$$

$$\Rightarrow A+B=0$$

$$-A+2A+4B+C=1$$

$$-2A+4B-C=1$$

$$\Rightarrow A = -\frac{2}{9} \quad B = \frac{2}{9} \quad C = \frac{1}{3}$$

$$\Rightarrow H(s) = -\frac{2}{9} \cdot \frac{1}{s+2} + \frac{2}{9} \cdot \frac{1}{s-1} - \frac{1}{3} \frac{1}{(s+2)^2}$$

$$\Rightarrow y_h(t) = -\frac{2}{9} e^{-2t} u(t) + \frac{2}{9} e^t u(t) - \frac{1}{3} t \cdot e^{-2t} u(t)$$

$\Rightarrow$ Hệ ~~phản~~ nhàn quá:

Hệ ổn định vì $\operatorname{Re}(s)$ lúc âm lúc dương

Bài 4.8. Cho hệ LTI cho bởi PVP.

$$\frac{d^2 y(t)}{dt^2} + \frac{dy(t)}{dt} + \frac{y(t)}{2} = 2 \frac{dx(t)}{dt} + x(t)$$

a) Hệ ổn định hay không?

b) Xét đáp ứng xung

c) Xét đáp ứng với đầu vào $x(t) = e^{-\frac{t}{2}} u(t)$.

Giải

a) Biến đổi Laplace 2 vế $\Rightarrow$

$$s^2 Y(s) + sY(s) + \frac{1}{2}Y(s) = 2sX(s) + X(s)$$

$$\Rightarrow (s^2 + s + \frac{1}{2}) Y(s) = (2s+1) X(s)$$

$$\Rightarrow H(s) = \frac{Y(s)}{X(s)} = \frac{2s+1}{s^2 + s + \frac{1}{2}}$$

Đ cực của hệ thống $s = -\frac{1}{2} \pm \frac{1}{2}i$ $\operatorname{Re}(s) < 0$.

$$s = -\frac{1}{2} - \frac{1}{2}i$$

$\Rightarrow$ Hệ thống ổn định

$$1) H(s) = \frac{2s+1}{s^2 + s + \frac{1}{2}} = \frac{2(s + \frac{1}{2})}{(s + \frac{1}{2})^2 + \frac{1}{4}} = 2 \cdot \frac{s + \frac{1}{2}}{(s + \frac{1}{2})^2 + \frac{1}{4}}$$

$$h(t) = \mathcal{L}^{-1}[H(s)] = \mathcal{L}^{-1} \left[2 \cdot \frac{(s + \frac{1}{2})}{(s + \frac{1}{2})^2 + \frac{1}{4}} \right] = 2 \left(e^{-\frac{1}{2}t} \cos \frac{1}{2}t \right) u(t)$$

$$= \left(e^{-\frac{1-j}{2}t} + e^{-\frac{1+j}{2}t} \right) u(t) = e^{-\frac{1}{2}t} \left(e^{j\frac{1}{2}t} + e^{-j\frac{1}{2}t} \right) u(t)$$

$$e) x(t) = e^{-\frac{t}{2}} u(t)$$

$$\Rightarrow X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt$$

$$= \int_0^{+\infty} e^{-t/2} e^{-st} dt = \int_0^{+\infty} e^{-t(s+\frac{1}{2})} dt$$

$$= \frac{1}{s+\frac{1}{2}} e^{-t(s+\frac{1}{2})} \Big|_0^{+\infty} = \frac{1}{s+\frac{1}{2}}$$

$$\Rightarrow Y(s) = H(s) X(s) = \frac{2}{2s+1}$$

$$\Rightarrow Y(s) = \frac{2(2s+1)}{(2s+1)(s^2+s+\frac{1}{2})}$$

$$= \frac{2}{s^2+s+\frac{1}{2}} = \frac{2}{(s+\frac{1-j}{2})(s+\frac{1+j}{2})}$$

$$= 2 \cdot \frac{-j}{s+\frac{1-j}{2}} + \frac{j}{s+\frac{1+j}{2}}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}[Y(s)] = 2 \cdot \left(-j e^{-\frac{1-j}{2}t} + j e^{-\frac{1+j}{2}t} \right)$$

C2: b. $H(s) = \frac{2s+1}{s^2+s+\frac{1}{2}} = \frac{2s+1}{(s+\frac{1-j}{2})(s+\frac{1+j}{2})}$

$$= \frac{1}{s+\frac{1-j}{2}} + \frac{1}{s+\frac{1+j}{2}}$$

$$\Rightarrow h(t) = \mathcal{L}^{-1}[H(s)] = \left(e^{-\frac{1-j}{2}t} + e^{-\frac{1+j}{2}t} \right) u(t)$$